GATE CS/DA Assignment Submission: Algorithms & Complexity

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Problem Statement:
State the Master Theorem formulation and derive the asymptotic tight bound for the divide-and-conquer recurrence relation:

$$T(n) = 2T\left(\frac{n}{2}\right) + \Theta(n), \quad \text{with } T(1) = \Theta(1)$$

Student Solution Derivation:
Step 1: Standard Master Theorem Form
The standard divide-and-conquer recurrence is represented as:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

where $a \geq 1$ represents the number of recursive subproblems, $b > 1$ represents the subproblem size reduction factor, and $f(n)$ is the non-recursive recombination work.

Step 2: Parameter Extraction & Critical Exponent Calculation
From the given recurrence $T(n) = 2T(n/2) + \Theta(n)$:

$$a = 2, \quad b = 2, \quad f(n) = \Theta(n)$$

We evaluate the critical exponent value $\log_b a$:

$$c_{\text{crit}} = \log_b a = \log_2 2 = 1 \implies n^{\log_b a} = n^1 = n$$

Step 3: Case Identification
We compare $f(n)$ with $n^{\log_b a}$:

$$f(n) = \Theta(n) = \Theta\left(n^{\log_b a}\right)$$

Since $f(n) = \Theta(n^{\log_b a})$, this problem strictly satisfies **Case 2** of the Master Theorem with $k = 0$.

Step 4: Asymptotic Tight Bound Conclusion
According to Case 2 of the Master Theorem:

$$T(n) = \Theta\left(n^{\log_b a} \log^{k+1} n\right) = \Theta(n \log n)$$

Final Answer: The recurrence has a tight complexity of $\Theta(n \log n)$ (as observed in MergeSort).